



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

ACTIVITY WORKSHEETS

For

Effective Project Management for Small Projects

TGS Ref No: TGS-2020505545

7 activities · one running case study: HarbourFront Logistics

Version v18.0 · 19 August 2026



ACTIVITY 1 WORKSHEET

Authorise the HarbourHub Click & Collect Pilot

Effective Project Management for Small Projects · TGS-2020505545 · Topic 1 · LO1 · 75 minutes

TEAM MEMBERS

DATE

TOOL

<https://alfredang.github.io/designthinking/>

ACTIVITY FOLDER

activities/activity-01-*/

THE SITUATION

HarbourHub Fresh has approved discovery work for a click-and-collect pilot at three outlets, but the sponsor has not yet authorised delivery. Operations wants speed, Finance wants a hard S\$180,000 ceiling, store managers fear disruption, and customers want reliable collection times. Your team must decide whether this is a viable small project and define the authority to proceed.

Record your team's output below.

1. Small-project qualification test

2. Primary user and evidence-based need

3. Cost–benefit outputs and sensitivity

4. Recommended option and rationale

5. SMART success measures

6. In-scope deliverables

7. Explicit exclusions

8. Milestones and budget cap

9. Sponsor, PM and core team

10. Assumptions, constraints and top risks

11. Kickoff agenda

12. Sponsor gate decision and conditions

SELF-CHECK BEFORE YOU FINISH

- ☐ The charter proves the work qualifies as a small project, names one accountable sponsor, contains quantified success measures and exclusions, stays within five months and S\$180,000, and carries an explicit authorisation decision.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.



ACTIVITY 2 WORKSHEET

Build the Integrated Scope, Schedule and Cost Baseline

Effective Project Management for Small Projects · TGS-2020505545 · Topic 2 · LO2 · 120 minutes

TEAM MEMBERS

DATE

TOOL

<https://lms-tms.tertiaryinfotech.com/>

ACTIVITY FOLDER

activities/activity-02-*/

THE SITUATION

The sponsor approved the charter with one condition: the three outlets must launch before Lunar New Year trading. You have high-level deliverables but no traceable requirements, work breakdown, dependency logic or time-phased budget. A vendor has proposed a date based only on optimism.

Record your team's output below.

1. Planning assumptions

2. Requirements and sources

3. MoSCoW decision

4. WBS and 100% rule check

5. WBS dictionary entries

6. Dependency logic

7. Forward/backward-pass results

8. Critical path and float

9. Gantt milestones

10. Basis of estimates

11. Time-phased cost baseline and reserves

12. Integration cross-check and repair

SELF-CHECK BEFORE YOU FINISH

- ☐ Every requirement traces to acceptance evidence and a WBS deliverable; the network produces a defensible critical path; the Gantt fits the five-month limit; and all scheduled work is funded within S\$180,000.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.



ACTIVITY 3 WORKSHEET

Diagnose and Prioritise the Pilot's Quality Failure

Effective Project Management for Small Projects · TGS-2020505545 · Topic 3 · LO3 · 75 minutes

TEAM MEMBERS

DATE

TOOL

<https://alfredang.github.io/fishbone/>

ACTIVITY FOLDER

activities/activity-03-*/

THE SITUATION

During outlet testing, only 82% of orders are ready at the promised collection time and pick accuracy has fallen to 94%. The software vendor blames store staff; store staff blame late inventory updates. The sponsor wants action before the team has separated symptoms, causes and normal variation.

Record your team's output below.

1. Reconciled check sheet

2. Fishbone effect and cause families

3. Facts versus hypotheses

4. Five-whys causal chain

5. Backwards causal check

6. Pareto vital few

7. Control-chart interpretation

8. SMART experiment

9. Owner, measure and review date

10. Sponsor explanation

SELF-CHECK BEFORE YOU FINISH

- ☐ The data total reconciles; Pareto reaches 100%; the five-whys chain ends at an actionable cause; the control chart interpretation matches the pattern; and the corrective experiment has an owner, measure and review date.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.



ACTIVITY 4 WORKSHEET

Build the Risk Register and Fund the Right Responses

Effective Project Management for Small Projects · TGS-2020505545 · Topic 4 · LO4 · 75 minutes

TEAM MEMBERS

DATE

TOOL

<https://lms-tms.tertiaryinfotech.com/>

ACTIVITY FOLDER

activities/activity-04-*/

THE SITUATION

The pilot has entered build. The POS interface may be delivered late, refrigerated stock counts are unreliable at one outlet, a key trainer is available for only two weeks, and a faster payment option could improve adoption but costs more. The remaining contingency reserve is S\$18,000.

Record your team's output below.

1. Risk versus issue decisions

2. Cause-event-impact statements

3. Probability and impact rationale

4. Priority exposure

5. Risk and response owners

6. Triggers

7. Preventive, contingency and fallback actions

8. Decision-tree EMV

9. Tornado sensitivity reading

10. S\$18,000 reserve allocation

11. Residual and secondary risks

SELF-CHECK BEFORE YOU FINISH

- ☐ Every priority risk has cause-event-impact wording, an owner, trigger, funded response and residual score; the decision-tree arithmetic reconciles; and the named contingency allocations total no more than S\$18,000.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.



ACTIVITY 5 WORKSHEET

Align Stakeholders with a Map, RACI and Engagement Plan

Effective Project Management for Small Projects · TGS-2020505545 · Topic 5 · LO5 · 75 minutes

TEAM MEMBERS

DATE

TOOL

<https://alfredang.github.io/raci/>

ACTIVITY FOLDER

activities/activity-05-*/

THE SITUATION

Two weeks before outlet testing, the POS vendor and operations manager each believe the other owns interface acceptance. Store managers receive conflicting instructions, Finance has not approved the payment milestone, and a high-influence outlet supervisor is quietly resisting the pilot.

Record your team's output below.

1. Stakeholder register

2. Power-interest map and evidence

3. Salience comparison

4. RACI matrix

5. RACI diagnostic corrections

6. Current versus desired engagement

7. Communication and feedback rhythm

8. Escalation brief

9. Role-play decision and updates

SELF-CHECK BEFORE YOU FINISH

- ☐ All material stakeholders are analysed; every critical deliverable has exactly one Accountable owner and at least one Responsible party; engagement gaps have two-way actions; and the escalation asks for a clear decision.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.



ACTIVITY 6 WORKSHEET

Deploy the Team and Resolve the Resource Bottleneck

Effective Project Management for Small Projects · TGS-2020505545 · Topic 6 · LO6 · 60 minutes

TEAM MEMBERS

DATE

TOOL

<https://alfredang.github.io/scrum/>

ACTIVITY FOLDER

activities/activity-06-*/

THE SITUATION

The test plan needs 22 person-days in the next two weeks, but only 16 are available. The same business analyst is assigned to testing, training materials and vendor clarification. The sponsor refuses a launch delay unless the team proves no lower-cost option exists.

Record your team's output below.

1. Organisation and escalation map

2. Available capacity by person and skill

3. Demand-capacity gap

4. Constraint

5. Kanban and WIP limits

6. Smoothing option

7. Levelling option

8. Crashing option

9. Fast-tracking option

10. Budget impact

11. Sponsor recommendation

12. Team handoff and conflict rules

SELF-CHECK BEFORE YOU FINISH

- ☐ The plan reconciles named work with actual capacity; the bottleneck is visible; each optimisation option has schedule, cost and risk consequences; and the chosen deployment fits the approved project constraints.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.



ACTIVITY 7 WORKSHEET

Control the Pilot, Decide the Change and Close the Project

Effective Project Management for Small Projects · TGS-2020505545 · Topic 7 · LO7 · 105 minutes

TEAM MEMBERS

DATE

TOOL

<https://alfredang.github.io/scrum/>

ACTIVITY FOLDER

activities/activity-07-*/

THE SITUATION

At the end of week 12, the baseline planned value is S\$108,000. Completed deliverables represent S\$90,000 of budgeted value and actual cost is S\$105,000. The sponsor also requests loyalty points, estimated at S\$24,000 and three weeks. Launch evidence is incomplete and the vendor wants final payment.

Record your team's output below.

1. Current Kanban and acceptance evidence

2. Burndown interpretation

3. PV, EV and AC

4. SV, CV, SPI and CPI

5. EAC and ETC forecast

6. One-page status report

7. Change impact analysis

8. Change decision and rationale

9. Updated baselines or decision log

10. Corrective action

11. Quality control interpretation

12. Acceptance and handover

13. Lessons learned

14. Closure recommendation

SELF-CHECK BEFORE YOU FINISH

- ☐ Earned-value calculations and forecast reconcile; the change decision states complete impacts; corrective action has an owner and due date; and closure is supported by acceptance, handover, contract and lessons evidence.
- ☐ All calculations reconcile to the supplied data and approved constraints.
- ☐ Every team member can explain the recommendation and trade-off.
- ☐ Completed editable tools or exports are saved in this folder.